Architecture Governance

Module 9

Module Objectives

- Architecture Governance
- The main components that make up an Architecture Governance Framework
- The TOGAF Architecture Governance Framework
- Architecture Governance in Practice
- Why Architecture Governance is beneficial
- Guidelines for establishing an EA Capability

Introduction to Governance

- Governance is the practice by which enterprise architectures are **managed** and **controlled**
- This includes
  - controls on the creation and monitoring of components and activities – ensuring introduction, implementation, and evolution of architectures
  - ensuring **compliance** with internal and external standards and regulatory obligations
  - supporting management of the above
  - ensuring **accountability** to external and internal stakeholders

Governance and the ADM

- Governance should be established in the **Preliminary Phase**
  - Usually an adaptation of existing governance and support models
- The **Architecture Board** should ensure that the ADM is being applied correctly
  - Compliance to the ADM is fundamental to the governance of the Architecture
- Governance plays a key role in **Phases G and H**
  - The implementation and then change management activities

Nature of Governance

- Governance ensures business is conducted properly
- It is about **effective** and **equitable usage** of resources to ensure sustainability of strategic objectives
- Basic principles of corporate governance
  - Focus on the **rights, roles** and equitable treatment of shareholders
  - Disclosure of transparency
  - **Accountability** of the Board to the stakeholders
  - Responsibilities of the board
    - **Reviewing** and guiding corporate strategy
    - Setting and monitoring management's performance objectives

Characteristics of Governance

- **Discipline**
- Transparency
- **Independence**
- Accountability
- Responsibility
- **Fairness**

Governance – Basic Principles

Governance is the system by which business corporations are **directed** and **controlled**.

The corporate governance structure specifies the distribution of rights and responsibility among different participants.

It spells out the **rules** and **procedures** for making decisions on corporate affairs.

Levels of Governance

- The hierarchy of governance domains includes
  - Technology Governance
  - IT Governance
  - Architecture Governance

- Each domain may exist at multiple geographic levels
  - Global
  - Regional
  - Locals

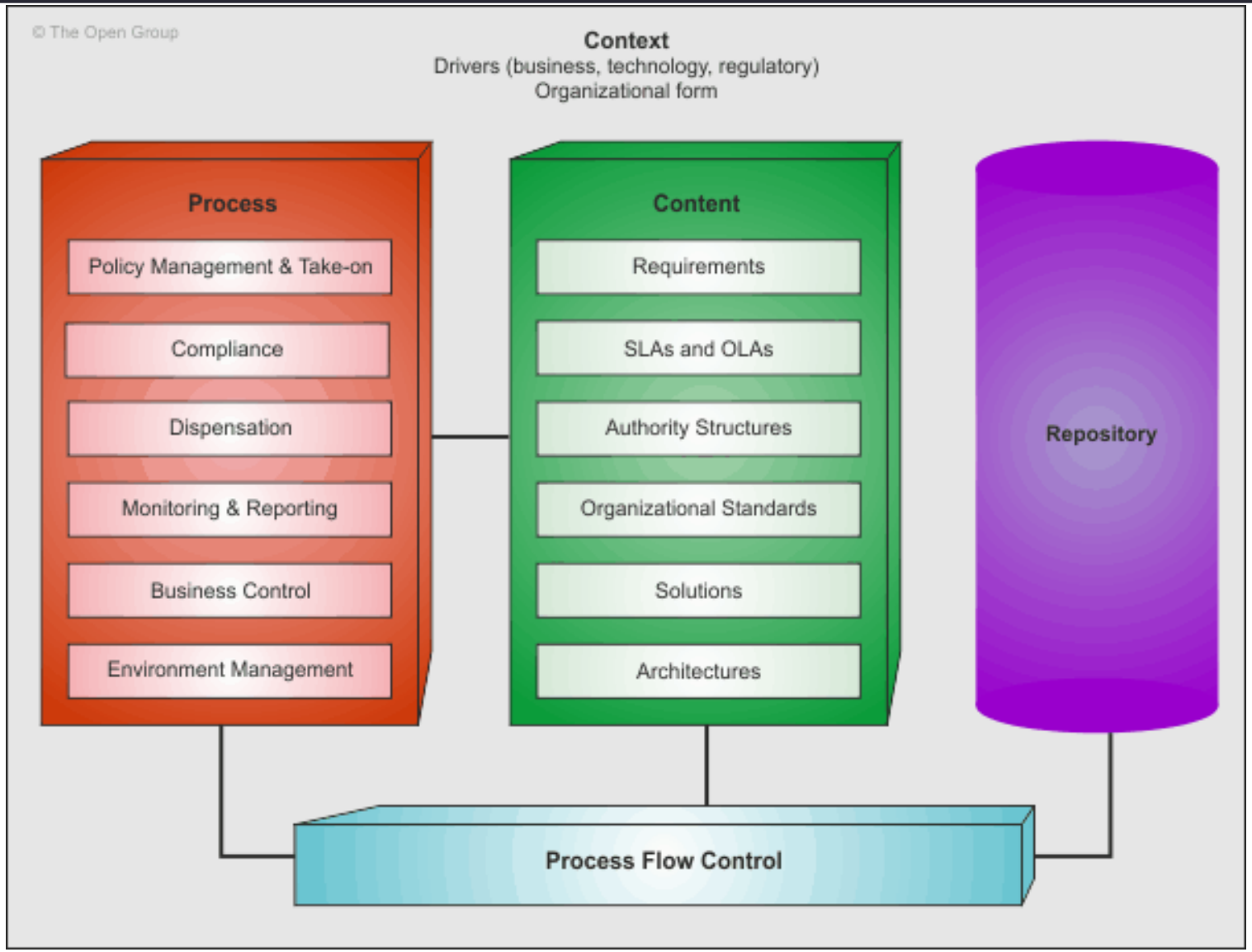
An IT Governance Framework - COBIT

- **COBIT** is an open standard of control of IT
- It was developed and promoted by the IT Governance Institute
- COBIT provides a generally accepted standard for good IT security and control practices
- There is also a set of Management Guidelines for COBIT, including **Maturity Models**, **Critical Success Factors**, **Key Goal Indicators**, and **Key Performance Indicators**
- The framework can help managers to control and measure IT resources

TOGAF Architecture Governance Framework

- **Phase G** of the TOGAF ADM is about **Implementation Governance** –the realization of architecture through change projects
- Architecture Governance covers management and control of all aspects of the development and evolution of enterprise architectures
- The Architecture Governance Framework is **generic** and can be adapted to an existing governance environment. It helps to identify effective processes and organizational structures, so that the **business responsibilities** can be described, communicated and managed

Conceptual Structure



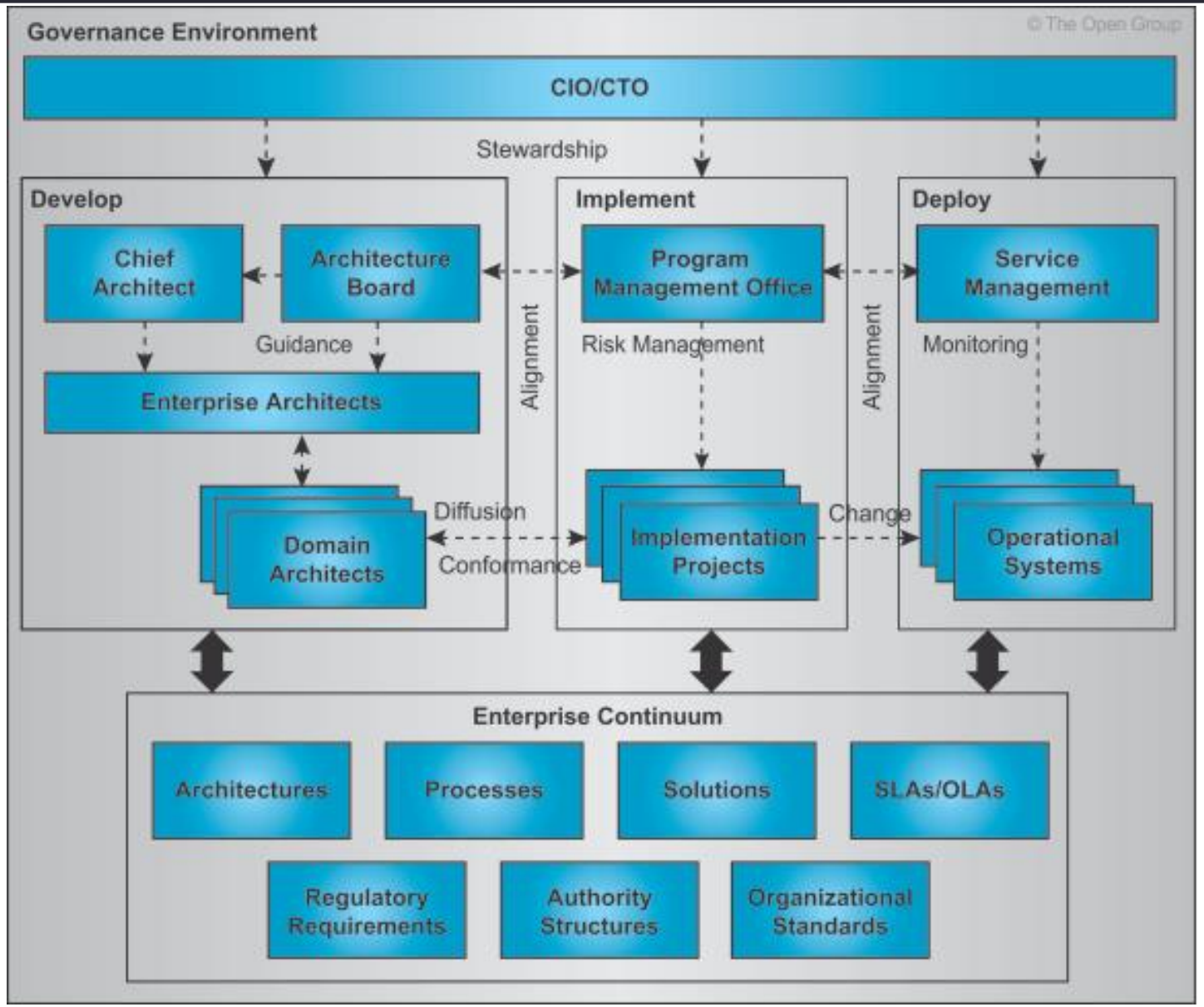
Conceptual Structure

- Architecture Governance is an approach, a series of processes, a cultural orientation and a set of responsibilities that **ensure the integrity** and effectiveness of architectures
- The split of **process**, **content** and **context** is key to supporting an architecture governance initiative – allows introduction of new governance material without impacting the processes and ensures framework flexibility

Conceptual Structure

- **The Architecture Governance Framework** is integral to the Enterprise Continuum, and **manages all content** for both the architecture and the architecture governance processes

Organizational Structure



Organizational Structure

- Governance is the **management** and **control** of the architectures
- To ensure effective control, it is necessary to have the **correct organizational structures** to support all governance activities
- Effective implementation requires IT governance processes, organizational structures, and capabilities including e.g.
  - Global and local governance board
  - Design authorities and Working parties

Benefits of Architecture Governance

- Links processes, resources, and information to **organizational strategies** and objectives
- Integrates and institutionalizes **best practices**
- Aligns with industry frameworks
- Enables organization to take full advantage of its assets
- Protects the underlying digital assets of the organization
- Supports **regulatory** and best practice **requirements**
- Promotes visible **risk management**

Architecture Governance in Practice

Key success factors include

- **Best practices** for submission, adoption, reuse, reporting, and retirement of architecture policies, procedures, roles, skills, organizational structures, and support services
- **Organizational responsibilities** and **structures** to support the architecture governance processes and reporting requirements

Architecture Governance in Practice

- **Tools** and **processes** to procedurally and culturally promote take-up
- Management of criteria to control architecture governance processes, **dispensations**, **compliance assessments**, SLAs, and OLAs
- Meet internal and external requirements for effectiveness, efficiency, confidentiality, integrity, availability, compliance, and reliability of architecture governance-related information, services, and processes

Architecture Board

- The Board **oversees implementation** of the governance strategy
- Board comprises of **representative stakeholders** responsible for review and maintenance of architecture typically at 2 levels
  - **Local** (**domain experts**, line responsibility)
  - **Global** (**organization-wide** responsibility)
  - Board has identifiable and articulated
    - Responsibilities and **decision-making** capabilities
    - Remit and authority limits

Architecture Board Operations

- TOGAF provides **guidance on operations** of the Board
- These are primarily focused on best practice for **meeting management**
- For example:
  - Meetings should be conducted with clearly defined agendas
  - Each participant attending a meeting should be fully prepared
- TOGAF provides a sample outline agenda

Architecture Contracts

- **Joint agreements** between development partners and sponsors on the deliverables, qualify and fitness-for purpose of an architecture

- Type of contracts
  - **Statement of Architecture Work**
  - Architecture Design and Development Contracts
  - Business Users' Architecture Contract

Architecture Contracts

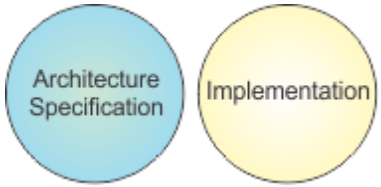
Use of Architecture Contracts ensures

- **Continuous monitoring** to check integrity, changes, decision-making, and audit of all architecture activities
- **Adherence to the principles**, standards, and requirements of the existing or developing architectures
- Identification of **risks**
- A set of **process** and **practices** that ensure accountability, responsibility, and discipline with regard to the development and usage of all architecture artifacts
- A **formal understanding** of the governance organization

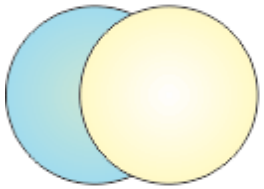
Architecture Contracts and the ADM

- Statement of Architecture Work created in **Phase A**
- Architecture Domains (Business, Data, Application, Technology)
- **Phase G**
- Implementation projects

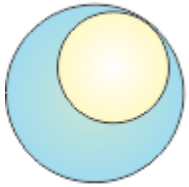
Architecture Compliance Terminology



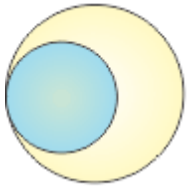
Irrelevant:
The implementation has no features in common with the architecture specification (so the question of conformance does not arise).



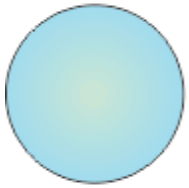
Consistent:
The implementation has some features in common with the architecture specification, and those common features are implemented in accordance with the specification. However, some features in the architecture specification are not implemented, and the implementation has other features that are not covered by the specification.



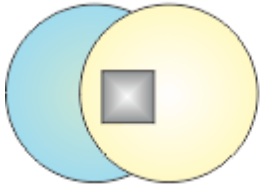
Compliant:
Some features in the architecture specification are not implemented, but all features implemented are covered by the specification, and in accordance with it.



Conformant:
All the features in the architecture specification are implemented in accordance with the specification, but some more features are implemented that are not in accordance with it.



Fully Conformant:
There is full correspondence between architecture specification and implementation. All specified features are implemented in accordance with the specification, and there are no features implemented that are not covered by the specification.



Non-conformant:
Any of the above in which some features in the architecture specification are implemented not in accordance with the specification.

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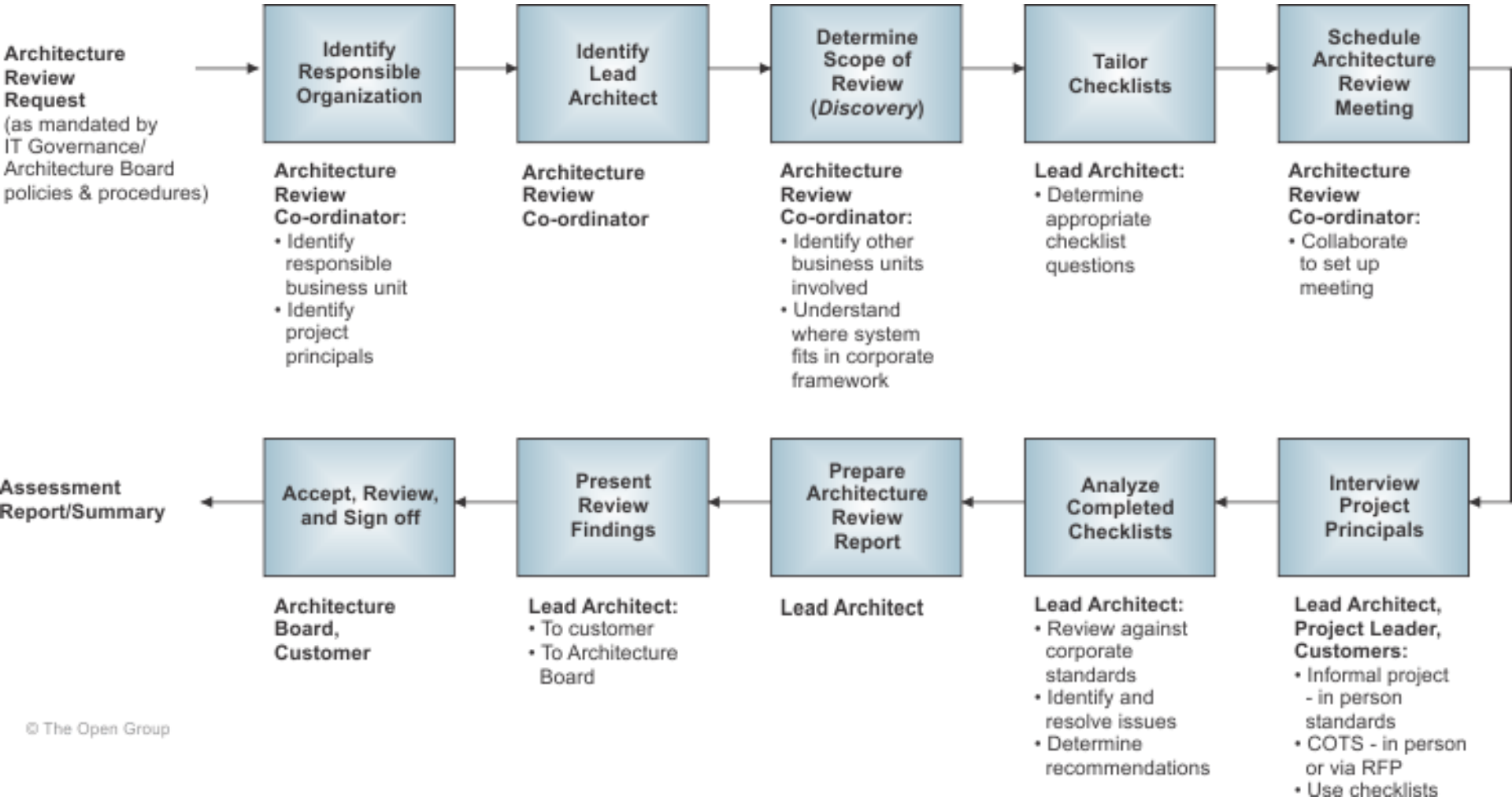
Architecture Compliance

- Two processes re defined to ensure compliance of projects with the Enterprise Architecture
  - Prepare **Project Impact Assessments** – project specific views that illustrate how the Enterprise Architecture impact a project
  - Perform an **Architecture Compliance Review**

Architecture Compliance Reviews

- **Catch errors** in the project architecture early
- Ensure the **best practices** to architecture work
- Provide an overview of the **compliance** to mandated standards
- Identify where the standards may **require modification**
- Identify services that are currently application-specific but might be provided as part of the enterprise infrastructure
- Document **strategies for collaboration**, resource sharing, and other synergies across multiple architecture teams
- Take advantage of **advances in technology**
- Communicate to management the **status of technical readiness** of the project
- Identify key criteria for **procurement activities**
- Identify and communicate significant **architectural gaps** to product and service providers

Architecture Compliance Review Process



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Establishing an Architecture Capability

- Use of the **ADM**
- Treat as on **ongoing practice**
- Address the four domain architectures
  - **Business** : the architecture **governance**, processes, organizational structure, requirements, products, etc.
  - **Data** : the structure of the organization's Enterprise Continuum and **Architecture Repository**
  - **Application** : the functionality and/or application services required to enable the architecture practice
  - **Technology** : **infrastructure requirements** and deployment in support of the architecture applications and **Enterprise Continuum**

Summary

Architecture Governance is the practice and orientation by which enterprise architectures and other architectures are **managed** and **controlled** at an enterprise-wide level

- Implementing a system of controls over the creation and monitoring of all architecture components and activities, to **ensure the effective introduction**, implementation, and evolution of architectures within the organization
- Implementing a system to **ensure compliance** with internal and external standards and regulatory obligations
- Establishing processes that support **effective management** of these processes
- Developing practices that **ensure accountability** to identified stakeholders, inside and outside the organization